

Task 3 — Tool Design: JurisdictAI Monitor

Option C — Analytical Design with Quantitative Component

One-Page Plain Language Summary

What the tool does:

JurisdictAI Monitor is a Python-based analytical tool that tracks whether a bank's credit-decisioning AI model is performing fairly and stably — and reports that performance differently depending on which country's rules apply. For a US bank, it watches for statistical model drift (PSI). For an EU deployment, it runs mandatory bias tests and checks whether human override mechanisms are documented. A single model deployed in both jurisdictions generates two separate compliance reports, because the rules are genuinely different.

The core problem it solves:

Banks deploying AI models in both the US and EU currently manage compliance with two separate spreadsheet-and-memo processes that have no common data layer. JurisdictAI Monitor unifies the monitoring data while keeping compliance outputs jurisdiction-specific. The alternative — treating both jurisdictions as if they have the same requirements — either over-engineers US compliance (expensive, unnecessary) or under-engineers EU compliance (illegal under AI Act).

What it does NOT do:

- Does not remediate model problems — detects and reports them only
- Does not cover GenAI models (EU AI Act Title III GenAI obligations require separate architecture)
- Does not automatically update when regulations change — a human specialist must update jurisdiction config files
- Is a monitoring and reporting aid, not a compliance guarantee — legal review remains required

1. Jurisdiction Configuration Layer

The tool stores regulatory parameters in versioned configuration files (YAML format in production, Python dictionaries in this implementation), one per jurisdiction. This allows parameters to be updated independently of the codebase.

Parameter	US — OCC 2026-13	EU — AI Act 2024	Significance
PSI revalidation threshold	> 0.25 = RED (revalidate)	No PSI standard	OCC explicitly references PSI; EU uses holistic monitoring
Fairness testing	Advisory (ECOA separate)	Mandatory (Article 10)	EU treats bias testing as legal obligation, not best practice
Fairness threshold (DPD)	None mandated	< 0.05 (BaFin / EU AI Office)	Informal EU standard; no equivalent US numerical threshold
Human oversight	Recommended	Mandatory (Article 14)	EU treats AI credit decisions as high-stakes by default
Pre-deployment check	Not required	Conformity assessment required (Art. 43)	EU creates pre-market gate; US does not
Incident reporting	Internal escalation	Notify national authority (Art. 73)	EU creates regulatory notification obligation
GenAI scope	Explicitly excluded	Included (Title III)	Major gap: OCC 2026-13 does not govern LLM-based decisions

2. Monitoring Metrics

US Jurisdiction (OCC 2026-13)

- Population Stability Index (PSI) — measures distribution shift between training data and current production data
- KS Statistic — measures model discrimination stability
- AUC-ROC trend — measures overall model predictive power over time

EU Jurisdiction (EU AI Act)

- Demographic Parity Difference (DPD) — measures approval rate disparity between protected and non-protected groups
- Equalized Odds Difference — measures false positive/negative rate disparity across groups
- Counterfactual Fairness Index (advisory, not regulatory) — tests whether decisions change under protected attribute counterfactuals

Both Jurisdictions

- Model accuracy (AUC-ROC) — fundamental performance baseline
- Feature importance stability — flags if model is implicitly relying on proxy variables for protected attributes

3. Failure Mode Analysis

Failure Mode	Impact	Detection Method	Mitigation
Rule change mid-period	Monitoring applies outdated thresholds	Config version audit log	Human-triggered config update; changelog required
Model drift	Fairness/performance metrics degrade silently	PSI (US) / rolling DPD (EU)	Automated alerts at yellow threshold before red
Jurisdictional misconfiguration	Wrong rules applied to deployment	Config validation on startup + confirmation dialog	Cannot fully automate; requires human confirmation
Two jurisdictions contradict	EU requires human oversight; US does not	Conflict detection in config layer	Surface conflict explicitly; do not silently resolve
Applicant pool distributional shift	Historical baselines unreliable	Covariate shift detection on feature distributions	Baseline recalibration procedure
Proxy variable emergence	Model re-learns protected attribute via correlates	SHAP feature importance stability monitoring	Alert when feature importance ranking shifts

4. Model Card (Governance Documentation Stub)

Field	Value
Tool name	JurisdictionAI Monitor v0.1
Tool type	Compliance monitoring and reporting (not a predictive model)
Intended use	Monitoring deployed credit-decisioning AI models for regulatory compliance across US (OCC 2026-13) and EU (AI Act 2024)
Out-of-scope use	Should NOT be used as sole basis for concluding regulatory compliance. Legal review required.
Data inputs	Production inference data provided by client bank; synthetic validation data used for threshold calibration
Known limitation: GenAI	Does not cover GenAI models (gap vs. EU AI Act Title III)
Known limitation: DPD threshold	0.05 is an informal standard, not a legally mandated number
Known limitation: counterfactual metric	Experimental; not validated against regulatory acceptability
Known limitation: config updates	Jurisdiction configs require manual update when regulations change

5. Quantitative Component Summary

Full implementation in Task3_Analysis.py. Key findings:

Metric	Value (Test Set)	US Compliance Status	EU Compliance Status
Model AUC-ROC	0.872	Above 0.70 benchmark	Adequate (no EU AUC standard)
Demographic Parity Difference (DPD)	0.385	Advisory only — no breach	FAIL — far exceeds 0.05 threshold
Equalized Odds Difference (EOD)	0.099	Advisory only	FAIL — exceeds 0.05 threshold
Population Stability Index (PSI)	0.034	GREEN — stable, no action	N/A — post-market plan applies instead

Key insight: The same model on identical data is compliant under US rules (OCC 2026-13) and non-compliant under EU rules (AI Act). Without jurisdiction-aware tooling, a compliance team reviewing only OCC outputs would consider this model acceptable for deployment.

6. What Was Not Included (Honest Scope Statement)

Gap	What Would Have Been Added	Why Not Included
SHAP/LIME explainability layer	Feature-level explanation for individual decisions; directly relevant to EU AI Act Article 13 and CFPB adverse action guidance	Requires a more complex model card and data pipeline; time constraint
Counterfactual fairness implementation	Causal graph-based individual fairness metric	Causal graph cannot be specified without domain knowledge of the specific model's design
Automated regulatory config updates	Rule update service with regulatory NLP parser	Production-grade feature; beyond assignment scope
GenAI governance layer	EU AI Act Title III foundation model obligations	Requires separate architecture; excluded from scope per OCC 2026-13 parallel decision
APAC jurisdiction (MAS 2024 AI MRM)	Singapore MAS guidance coverage alongside US/EU	Would extend scope beyond assignment requirement of two jurisdictions